Chunxiao (Calvin) Lin

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Research Summary

Hardware researcher building **FPGA accelerators for reservoir computing (Echo State Networks) with on-chip learning** for **Next-G wireless**: from cost-efficient neuron and readout microarchitectures (DSP48 arrays, systolic arrays, fixed-point RLS training) to **end-to-end real-time PC–FPGA testbeds** validated over the air and on a distributed-MIMO channel emulator. 12 publications, including first-author papers in IEEE TCAD, ICCAD, ICNC, ISQED and Frontiers.

Education

Virginia Tech, Ph.D. in Computer Engineering GPA: 4.0/4.0 Sep 2021 – May 2027 (Expected)

- Advisor: Prof. Yang (Cindy) Yi | Member of BRICCS@VT (Brain-Inspired Computing, Communication, and Security Lab)
- Focus: FPGA acceleration and on-chip learning of **Echo State Networks (ESNs)** for MIMO symbol detection and channel prediction

University of Southern California, Master of Engineering GPA: 3.95/4.0 Sep 2019 – May 2021

- Courses: Computer Architecture, VLSI Design, Digital System Design

Technical Skills

Hardware design: Verilog, SystemVerilog, RTL microarchitecture, DSP48 datapaths, systolic arrays, fixed-point arithmetic, timing and power analysis

FPGA platforms: Xilinx Virtex-7 (VC707, VC709), Zynq-7000 SoC, Zynq UltraScale+ (ZCU104), RFSoC (ZCU216)

Interfaces & systems: 10G/1G Ethernet (MAC, PCS/PMA), UDP/IP (lwIP), AXI, PS–PL co-design, USRP SDRs (B210, N310, X310), GNU Radio, Keysight PROPSIM channel emulator

Software & tools: Python, MATLAB, C; Xilinx Vivado, Vitis, Quartus, Cadence, ns-3

Research Experience

Graduate Research Assistant, BRICCS@VT, Virginia Tech Sep 2021 – Present

Real-Time PC–FPGA Testbeds for Distributed MIMO Channel Prediction Feb 2026 – Jul 2026

- Built two end-to-end testbeds that close a live 4×2 distributed-MIMO loop (4-antenna gNB, two 2-antenna UEs, USRP N310/X310, Keysight PROPSIM F16): a 1 GbE PS–PL hybrid (lwIP UDP/IP on the ARM, accelerator in the PL) and a 10 GbE pure-RTL design with the Ethernet MAC/PHY and two windowed-ESN predictors in programmable logic.
- Achieved $\leq$**0.9 ms end-to-end latency** and **1,250 channel predictions/s per UE** at 3.42 W on the 10 GbE testbed, lowering NMSE from 0.0119 (outdated CSI) to 0.0087 in live operation.

Hardware–Software Co-design of an On-chip ESN Apr 2025 – Sep 2025

- Implemented an on-chip ESN with PL-based inference and PS-based learning on a Xilinx Zynq-7000 SoC FPGA.

High-Speed Reservoir Neuron for MIMO Channel Prediction *ICNC 2026* Jun 2024 – Aug 2025

- Designed a hybrid DSP–CLB neuron (nine DSPs + carry-save adder tree) and a range-addressable tanh LUT: **+25% DSP efficiency**, 13% shorter processing time and 25% less tanh memory; the full ESN runs at **65.98 MS/s** and beats AR prediction by 5.7–14.2 dB in NMSE.

FPGA-based ESN with On-chip RLS Learning for MIMO Channel Prediction *IEEE TCAD 2026* Apr 2024 – Dec 2024

- Architected the first complete on-chip ESN with recursive-least-squares training on FPGA: a DSP48 systolic-array readout and an inversion-free, three-stage pipelined RLS engine whose latency grows only linearly with the state size.
- **3.87** μ**s** per training + inference step and 264 Mb/s at 1.2 W: $>$**1600**$\times$ **/ 290**$\times$ **the energy efficiency** of CPU / GPU implementations, and **5.3**$\times$ **lower NMSE** than the prior on-chip-trained FPGA ESN.

ESN Symbol Detection for (Massive) MIMO-OFDM on FPGA *ICCAD 2023, Frontiers 2024* Jun 2022 – Dec 2023

- Designed a nine-DSP48 reservoir neuron and built an SDR/FPGA platform (USRP, GNU Radio, Virtex-7) for over-the-air 2×2 MIMO-OFDM detection: **3.3**$\times$ **the throughput** of the prior SISO accelerator, hardware BER within 0.03% of software, and lower BER than MMSE in every scenario.
- Extended the ESN detector to massive MIMO, outperforming LMMSE from 2×2 up to 4×64 antennas with only 8–16 reservoir neurons.

Optimized Reservoir Node Architecture for FPGA-based RC Systems *ISQED 2022* Oct 2021 – Feb 2022

- Designed a bit-serial, CSD-encoded neuron that hard-wires the fixed sparse weights: **80.7% fewer LUTs** and DSPs cut from 162 to 16 versus a DSP-based ESN.

Other Projects

Low-Power 10T-SRAM for In-Memory Computing (USC) Oct 2020 – Dec 2020

- Designed a 64×64 10T-SRAM array with two asymmetric sense amplifiers for in-memory NOR/NAND; reduced leakage with high-Vt devices, transistor sizing, stacking and negative Vgs.

Dynamically Secured Scan Chain and Scan-based Attack (USC) *DFT 2021* Aug 2020 – Nov 2020

- Built a Python flow converting bench netlists to Verilog with dynamic encryption logic and verified the attack model in ModelSim; contributed to GF-Flush, which recovers 500-bit keys in under 7 s, over 4300× faster than SAT attacks.

Publications

- **C. Lin**, U. S. Khan, M. Zheng, M. F. Azmine, L. Liu, D. J. Jakubisin, and Y. Yi, "FPGA-based Reservoir Computing with On-chip Learning for MIMO Channel Prediction," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2026. doi: 10.1109/TCAD.2026.3715776.
- S. S. Chowdhury, M. R. Sarkar, J. Ling, **C. Lin**, and Y. Yi, "SPARC: Spiking Plasticity-Adaptive Reservoir Computing with On-Chip Reward-Modulated Triplet STDP for Spectrum Sensing," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 2026. *Submitted*.
- **C. Lin**, U. S. Khan, L. Liu, and Y. Yi, "FPGA-based High-speed Reservoir Computing for MIMO Channel Prediction," in *Proc. International Conference on Computing, Networking and Communications (ICNC)*, Maui, HI, USA, 2026, pp. 465–470. doi: 10.1109/ICNC68183.2026.11417007.
- A. Liu, M. F. Azmine, **C. Lin**, and Y. Yi, "Reservoir Computing: Foundations, Advances, and Challenges Toward Neuromorphic Intelligence," *AI*, vol. 7, no. 2, Art. no. 70, 2026. doi: 10.3390/ai7020070.
- R. Gaurav, S. Sinha, **C. Lin**, T. C. Stewart, L. Liu, and Y. Yi, "Multivariate Legendre-SNN on Loihi-2 for Time Series Classification and 5G Jamming Detection," *IEEE Transactions on Circuits and Systems for Artificial Intelligence (TCASAI)*, 2026. doi: 10.1109/TCASAI.2026.3658587.
- A. Liu, **C. Lin**, D. Rosen, and Y. Yi, "Echo State Networks in Reservoir Computing: Foundations, Benchmarks, and Applications to Next-G Wireless Communication," in *Proc. IEEE International Conference on Data Mining Workshops (ICDMW)*, Washington, DC, USA, 2025, pp. 3076–3080. doi: 10.1109/ICDMW69685.2025.00405.
- K. Yang, Y. Hu, D. Chen, **C. Lin**, Y. Yi, P. A. Beerel, and P. Nuzzo, "Machine Learning-Enhanced Analysis and Design for Trustworthy Integrated Circuits," in *AI-Enabled Electronic Circuit and System Design: From Ideation to Utilization*. Cham, Switzerland: Springer, 2025, pp. 467–496. doi: 10.1007/978-3-031-71436-8_13.
- **C. Lin** and Y. Yi, "Enhancing FPGA CAD Flow with AI-Powered Solutions," in *AI-Enabled Electronic Circuit and System Design*, A. Iranmanesh and H. Sayadi, Eds. Cham, Switzerland: Springer, 2025, pp. 225–256. doi: 10.1007/978-3-031-71436-8_7.
- **C. Lin**, M. F. Azmine, Y. Liang, and Y. Yi, "Leveraging Neuro-Inspired AI Accelerator for High-Speed Computing in 6G Networks," *Frontiers in Computational Neuroscience*, vol. 18, Art. no. 1345644, 2024. doi: 10.3389/fncom.2024.1345644.
- **C. Lin**, M. F. Azmine, and Y. Yi, "Accelerating Next-G Wireless Communications with FPGA-Based AI Accelerators," in *Proc. IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2023. doi: 10.1109/ICCAD57390.2023.10323918.
- **C. Lin**, Y. Liang, and Y. Yi, "FPGA-based Reservoir Computing with Optimized Reservoir Node Architecture," in *Proc. 23rd International Symposium on Quality Electronic Design (ISQED)*, Santa Clara, CA, USA, 2022, pp. 1–6. doi: 10.1109/ISQED54688.2022.9806247.
- D. Chen, **C. Lin**, and P. A. Beerel, "GF-Flush: A GF(2) Algebraic Attack on Dynamically Secured Scan Chains," in *Proc. IEEE International Symposium on Defect and Fault Tolerance in VLSI and Nanotechnology Systems (DFT)*, 2021, pp. 1–6. doi: 10.1109/DFT52944.2021.9568356.